

SOFTWARE ENGINEERING 381 - SEN381

CIVICCONNECT PROJECT

MILESTONE 1 (M1)

Engineering Foundation & Requirements Baseline

Team size	Exactly 3 students
Raw assessment	50 marks
Project weighting	15 project marks
Shared team artefacts/evidence	30 raw marks
Individual presentation	5 raw marks
Individual engineering defence	15 raw marks
Primary controlled output	PED v1.0 — Engineering Baseline
Governing document	SEN381 CivicConnect Master Project Brief

Milestone Gate

M1 establishes the controlled engineering foundation. It is not an architecture, technology-selection or implementation milestone. The team must demonstrate what is being engineered, for whom, within which constraints, and what must be anticipated before later consequential decisions are made.

1. Relationship to the Master Project Brief

This is a deliberately slim milestone brief. The SEN381 CivicConnect Master Project Brief is the single source of truth for all project-wide rules, standards, team formation and accountability, document quality, GitHub governance, two-reviewer approval, AI accountability, requirements and traceability standards, risk and decision controls, baseline management, evidence expectations, presentation standards and assessment principles

Apply, do not repeat

Where this milestone refers to a project-wide standard, students must locate and apply the relevant requirement in the Master Project Brief. A requirement is not removed merely because it is not repeated here.

1.1 Prior Learning Applied and SEN381 Progression

Milestone 1 deliberately applies knowledge developed in earlier modules. Students are not expected to relearn stakeholder analysis, requirements, programming or basic design concepts. Instead, they must use that prior knowledge as evidence within a controlled Software Engineering environment.

Prior learning applied

- **Systems Analysis & Design:** stakeholder identification and analysis; functional and non-functional requirements; scope and system boundaries; modelling; documentation; and analysis of competing needs.
- **Programming, Object-Oriented Programming, Web Programming and Databases:** technical understanding of implementable requirements; data and interface implications; validation; dependencies; security implications; and the effect of requirements on later construction.
- **Problem-solving and project foundations:** problem decomposition, prioritisation, teamwork, planning, constraints and reasoned solution development.

SEN381 progression

At NQF Level 8, the progression is from producing isolated analysis or implementation artefacts to engineering a controlled, traceable and defensible baseline. Requirements, risks, decisions, repository history and evidence must connect to later architecture, construction, verification, deployment, operation and controlled change. Students must therefore explain not only what an artefact contains, but why it exists, what evidence supports it, what trade-offs it creates and what downstream consequences may follow.

Acronym use: Acronyms and abbreviations used in this milestone follow the Acronyms and Abbreviations table in the SEN381 CivicConnect Master Project Brief. Important terms should still be written in full on first use where this improves clarity.

2. Milestone Objective

Analyse the CivicConnect problem within its stakeholder, scope, schedule, cost/resource, quality, security and risk constraints; establish clear, testable and traceable requirements; and create a controlled collaborative engineering baseline that anticipates later architecture, construction, verification, deployment, operation and change.

Central engineering question

What exactly are we committing to engineer, for whom, within what constraints, and what must we consider now to avoid unnecessarily constraining the project later?

3. Required Milestone Outputs

The following outputs must be integrated into PED v1.0 and/or linked as controlled project artefacts. Quality is assessed against the Master Project Brief. Document length is not an assessment objective.

Required output	M1 expectation	Evidence to present
Problem & business need	Concise CivicConnect-specific analysis of the problem and intended stakeholder/business value; do not rewrite the scenario.	PED section linked to stakeholder needs.
Stakeholder analysis	Identify relevant stakeholders, needs, influence/interest and meaningful conflicts or competing expectations.	Stakeholder artefact plus examples of requirements derived from it.
Scope baseline	Explicit in-scope, out-of-scope and deferred/future scope.	Controlled scope statement; defend at least one deliberate exclusion/deferment.
Requirements & acceptance criteria	Prioritised functional requirements and measurable non-functional requirements with unique IDs, sources and appropriate acceptance criteria.	PED requirements; demonstrate examples of testable FRs/NFRs.
Constraints	Record and analyse scope, schedule, cost/resources, quality and security constraints and their engineering implications.	Constraint evidence; explain at least one interaction/trade-off or ripple effect.
Initial RTM	Establish traceability from stakeholder/source to requirement and acceptance criteria, with structure for later design, change, test and release evidence.	Live RTM; trace at least one requirement end-to-end for the evidence currently available.
Initial Risk Register	Project-specific risks with probability, impact, priority, mitigation, contingency, owner and status.	Live register; defend the highest-priority risk.
Forward Engineering Considerations	Identify 5–7 later lifecycle concerns that require attention now, without prematurely implementing or deciding them.	Forward Engineering Considerations Register/map.
Engineering Decision Log	Create the controlled decision structure and record only genuine M1 decisions already made. Record justified deferment where evidence is not yet sufficient.	Decision log; defend one decision or deliberately deferred decision.
GitHub & team governance	Establish repository and team controls in accordance with the Master Project Brief, including protected main and the required review/approval model.	Live repository/settings/project evidence.
AI Usage Register	Establish the register and record material AI-assisted work and the human verification applied.	Live register plus verified example where AI was used.
PED v1.0 & baseline sign-off	Integrate the foundation into one coherent controlled PED, review it as a team and complete formal baseline sign-off/gate evidence.	PED v1.0, version history and baseline gate/sign-off evidence.

3.1 Progressive GitHub Evidence — Start from Milestone 1

GitHub evidence must be created progressively throughout the milestone. Students must not wait until presentation preparation to discover that repository history, issues, branches, Pull Requests, reviews, approvals and controlled artefact changes form part of the assessment evidence.

- Place appropriate controlled project artefacts in the team repository from the beginning of M1 and update them regularly.
- Use meaningful issues/tasks, commits, branches and Pull Requests for substantive changes to project artefacts, not only for application code.
- Apply the protected-main and required review/approval controls defined in the Master Project Brief from the start of controlled project work.
- Repository history must show authentic progression over time. Bulk uploads or reconstructed activity immediately before assessment do not demonstrate a controlled engineering process.
- Documentation, requirements, registers, decisions and other engineering artefacts can provide valid repository evidence before substantial application coding begins.

Key reminder: GitHub is an engineering control and evidence environment, not merely the place where final code is uploaded.

4. Forward Engineering Considerations

Assignment 1 introduced the idea that some decisions must be considered before they are implemented. M1 now applies that reasoning to CivicConnect. The team must identify 5–7 project-specific later concerns that already influence current requirements, constraints, risks or assumptions.

- Examples may include security, testability, automated build/integration, deployment environment, data migration/recovery, observability, scalability, maintainability or operational cost.
- Do not copy Assignment 1 research into M1 as a substitute for project-specific engineering evidence.
- For each concern, state why it matters now, what later decision/activity it may influence, what information is still missing, and the risk of ignoring it.
- It is acceptable—and often correct—to deliberately defer a decision when the team can explain what evidence is still required before deciding.

5. Explicit M1 Boundaries

M1 is a foundation milestone. Unless specifically approved as exploratory evidence, the following are not required M1 project decisions or deliverables:

- Final technology-stack selection.
- Final software architecture.
- Detailed database schema/persistence implementation.
- Detailed UI implementation.
- API implementation.
- Design-pattern implementation.
- CI pipeline implementation.
- Extensive application coding.
- Production deployment.

Thinking ahead is not premature implementation

Students may record future requirements, constraints, assumptions, risks and decision criteria for these areas. The purpose is to preserve later options and prepare evidence—not to race ahead of the lifecycle.

6. Assessment Structure — 50 Raw Marks

M1 is assessed on a 50-mark raw scale to support fair mark discrimination and is then converted to its approved project weighting of 15 marks. The raw scale does not change M1's weighted contribution to the project.

Assessment component	Raw marks	Purpose
Shared team artefacts & engineering evidence	30	Quality, control, traceability and readiness of the M1 engineering baseline.
Presentation Skill & Professional Communication	5	Individual professional engineering communication: structure, clarity, concise explanation, effective use of controlled evidence/visuals, delivery, timing, transitions and conduct.
Individual Engineering Defence & SE Understanding	15	Individual command of evidence, principles and standards, traceability, trade-offs, downstream consequences, accountability and quality of responses to assessor questioning.

Individual marks can differ

The project is collaborative, but competence is assessed individually. The 30 team-evidence marks are normally shared by the registered team; the 20 individual examination marks are awarded separately. Students in the same team may therefore receive substantially different M1 results.

6.1. Team Evidence Assessment — 30 Marks

Criterion	Marks	Assessment focus
Problem, Stakeholder Analysis & Business Value	3	Project-specific understanding, stakeholder needs/conflicts and value linkage.
Scope Baseline & Constraints	5	Controlled boundaries plus scope, schedule, cost/resources, quality and security constraints, including interactions/trade-offs.
Requirements & Acceptance Criteria	7	Prioritised, traceable and testable FRs/NFRs with measurable acceptance criteria.
RTM & Traceability Foundation	3	Stable identifiers and usable source → requirement → acceptance traceability

		prepared for later lifecycle evidence.
Risk & Forward Engineering Considerations	5	Prioritised risks plus credible forward thinking, downstream consequences and justified decision deferment.
GitHub Governance, Team Accountability & AI Controls	4	Operational project-wide controls, meaningful peer review/accountability and verified AI-use evidence.
PED Quality, Document Control & Baseline Sign-off	3	Coherent controlled PED v1.0, version control, team review and credible baseline gate evidence.

The detailed mark breakdown, partial-credit guidance, weak-evidence indicators and assessor probes are contained in the M1 Excel Rubric and Marking Guide.

6.2. Individual Examination — 20 Marks

6.2.1 Presentation Skill & Professional Communication — 5 Marks

Each student is assessed on professional engineering communication. Presentation skill is separate from technical understanding and does not replace engineering evidence.

- Logical structure and flow.
- Clarity and conciseness.
- Effective professional use of controlled artefacts and visual evidence.
- Delivery and engagement: explain rather than read.
- Timing, transitions/handover and professional conduct.

6.2.2 Individual Engineering Defence & SE Understanding — 15 Marks

- Command of actual project artefacts and evidence — 3 marks.
- Understanding of Software Engineering principles, standards and purpose — 3 marks.
- Traceability and relationships across artefacts — 3 marks.
- Forward-thinking, trade-offs and downstream consequence reasoning — 3 marks.
- Individual accountability, contribution and quality of responses — 3 marks.

Exam evidence

Assessors may use follow-up questions to distinguish rehearsed recall from genuine understanding. Any student may be asked about any controlled team artefact, regardless of who originally prepared or presented it.

7. Presentation Expectations

- Present engineering evidence, not a summary of the PED.
- Use live controlled artefacts where practical: PED, RTM, Risk Register, repository/settings/issues, decision log and AI Usage Register.

- Presentation and individual engineering defence are compulsory for every registered team member. Contribution to shared artefacts, documentation, analysis, coding, repository work or any other milestone activity does not replace the requirement to present and defend the team's engineering evidence.
- Slides may guide the presentation but must not replace controlled evidence.
- For each important artefact, be ready to show it, explain its Software Engineering purpose, connect it to other artefacts and explain its downstream consequence.
- Presentation quality and engineering defence are assessed separately. Strong delivery cannot compensate for weak evidence or weak understanding.
- The assessor may direct questions to any team member and use follow-up questions where necessary to test depth and authenticity.
- Recommended planning assumption: approximately 12–15 minutes team evidence presentation followed by individual questioning, subject to lecturer scheduling.

Absence / exceptional circumstances: A student who is unable to present because of illness or another formally accepted exceptional circumstance must communicate with the lecturer and make the necessary arrangements to present at a later approved date. Absence does not remove the presentation and individual defence requirement.

8. Indicative Defence Questions

Questions will be selected according to the team's actual evidence. These examples indicate the required depth and are not a script.

1. Which stakeholder expectation was hardest to reconcile, and how did it affect scope or requirements?
2. If the client adds a major feature after baseline without changing the deadline, what must the team analyse?
3. Show one NFR and explain how it could constrain an M2 architecture or technology choice.
4. Trace one requirement from source to acceptance criteria. What evidence will be added later?
5. Which risk deserves the most attention now, and why?
6. Why are deployment, automated testing or operations relevant in M1 when they are implemented later?
7. Which decision did your team intentionally not make, and what evidence is still needed?
8. Why must a substantive change entering main receive two approvals from team members other than the author?
9. Show one AI-assisted contribution. What did you verify, reject or change?
10. A baselined requirement changes tomorrow. Which artefacts and constraints must be revisited?
11. How will you later know CivicConnect succeeded beyond “the code works”?
12. Give one early shortcut that could create technical debt later.

9. Milestone Gate Decision

M1 concludes with a baseline-readiness decision. This engineering gate is recorded separately from the numeric mark.

Gate outcome	Meaning
ACCEPTED	PED v1.0 provides a sufficiently controlled foundation for the next project phase.
CONDITIONALLY ACCEPTED	Minor corrections/conditions are recorded; the team may proceed

	subject to those conditions.
REVISION REQUIRED	The foundation is not sufficiently controlled to justify consequential architecture/design decisions; required revisions must be completed.

9.1 Research Evidence, Project Evidence and Referencing

Where students rely on external sources, standards, professional guidance or research to support an engineering claim, the source must be cited appropriately. Research evidence and project evidence are not interchangeable.

- A citation may support why a practice, standard, quality attribute or engineering concern is relevant.
- A citation does not prove that the team implemented a control correctly; implementation claims require authentic project evidence such as repository history, tests, configuration, traceability or other controlled artefacts.
- Artificial Intelligence output is not a source and does not replace credible referencing or human verification.

10. Submission & Readiness Checklist

- PED v1.0 is complete, coherent, versioned and reviewed.
- Scope and requirements are baselined rather than merely drafted.
- RTM, Risk Register, Decision Log, Forward Engineering Considerations and AI Usage Register are current.
- GitHub governance required by the Master Project Brief is demonstrable.
- Team Working Agreement is agreed and controlled.
- Baseline sign-off/gate evidence is ready.
- All students can independently locate and explain the project foundation and its Software Engineering significance.
- All students are prepared to present professionally and defend the team's evidence individually.

Final M1 test

The team should be able to answer: What are we engineering, why, for whom, within what constraints, what have we formally committed to, what have we deliberately not decided yet, how will we control what changes next, and can every team member defend that foundation?

Academic Integrity and Referencing Disclaimer: All submitted work must include appropriate in-text citations and a corresponding reference list. Failure to provide in-text citations and references will result in an automatic mark of zero (0). Plagiarism, fabricated or misrepresented evidence, and inappropriate presentation of another person's or AI-generated work as the student's own are subject to the applicable academic-integrity requirements. Every registered student must present and complete the individual engineering defence; contribution to other milestone work is not a substitute for presentation.